

Revisiting Groundwater Recharge under Coupled Constraints

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Abstract

We examine the behaviour of soil carbon pools in urban heat islands. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting.

1. Introduction

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The measurements reported here were collected over twelve independent trials under controlled conditions. Calibration drift over the trial period remained below the threshold set out in the protocol.

Residual error grows approximately linearly with the number of coupled parameters. Neither of the alternative formulations improved on the simple additive model in our setting. Longer observation windows narrow the confidence interval but do not shift the central estimate. Sensitivity to the smoothing window proved modest across the range we examined. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Taken together, these findings argue for treating the two regimes separately in future analyses. Calibration drift over the trial period remained below the threshold set out in the protocol. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting.

2. Related Work

Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows approximately linearly with the number of coupled parameters. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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3. Materials and Methods

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4. Experimental Setup

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5. Results

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